

Date: November 7, 2025

LETTER OF PROJECT COMPLETION

This letter certifies that thefollowingstudents from **Thirdyear Electronicand Telecommunicationdepartment from VIT** has successfully completed a specialized technical project titled,

Project Title: LoRa-based Remote Control & Monitoring for Industrial Automation Applications

Students Names:

1) Shreyas Phansalkar
2) Pushpa
3) Raghvi Narayanan

This project was undertaken under the guidance of **Prof. Ashwinee Barbadekar** and in association with **Amotech Labs**, aimed at developing a real-time, computer vision–based automated quality control and sorting system using embedded image processing technology.

System Architecture – Designed a LoRa-based remote control and monitoring system using E220 and EBYTE E32 LoRa modules, supporting long-range, low-power industrial communication.
Multi-Controller Implementation – Implemented and validated the system on ESP32, PIC18F47K42, and PIC18F452, demonstrating controller-level portability and flexible system design.
Sensing & Data Acquisition – Integrated ADC channels and a temperature–humidity sensor on the field module to acquire real-time environmental and analog data for monitoring applications.
Relay Control Unit – Integrated relay outputs with the controller on the control module, enabling remote actuation of industrial loads based on commands received over LoRa.
Display Interface – Added a display module to visualize sensor values, ADC readings, and system status transmitted wirelessly from the field unit.
UART Communication – Configured and validated UART communication on PIC18F452 for reliable LoRa module interfacing and data exchange.
Reliable Wireless Communication – Established robust bidirectional LoRa communication for real-time data transmission and remote control over long distances.
Industrial Automation Focus – Delivered a scalable and modular solution suitable for remote monitoring, relay-based control, and data logging in industrial automation environments.
Outcome – Successfully demonstrated a complete end-to-end LoRa-based monitoring and control system, showcasing strong skills in embedded systems integration, wireless communication, and industrial automation system design.

We commend Shreyas Phansalkar, Pushpa, and Raghvi Narayanan on the successful completion and technical excellence demonstrated in this innovative embedded vision system.



Name: Amol Manal
Designation: Owner/ Embedded & Controls Engineer
Company Name: Amotech Labs, Pune
Email: amotechlabs@gmail.com **Contact No.:** 8329537565

Prof. Ashwinee Barbadekar
VIT Faculty Mentor

Contact Details:
(Whats App & Online Sales)
+91-8329537565
www.amotechlabs.com
Amol Manal (Prop.)
PG Cert. in Robotics and Automation - Conestoga College, Canada
www.linkedin.com/in/AmolManal
www.linkedin.com/company/amotech-labs
amotechlabs@gmail.com
Workplex, near Ashoka Centre
,Satara Road, Pune - 411 009.